

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 4 & Lecture 5 (Ch. 3)

The Goods Market

MICHELA BRAGA

WHERE WE ARE

Fundamentals

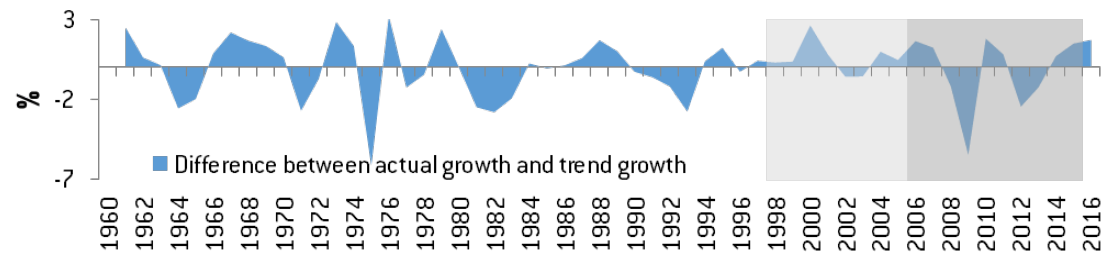
1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

THE BUSINESS CYCLE

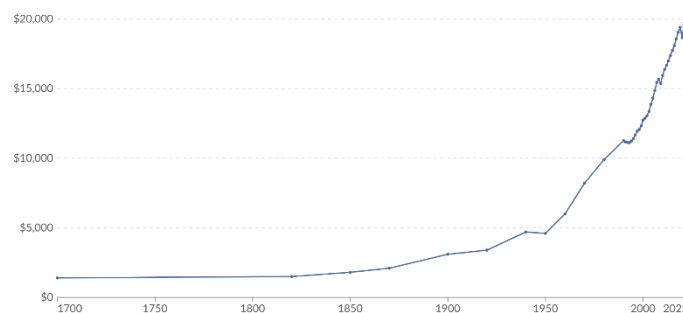
- Looking at the data we observe that real GDP:
 1. fluctuates around an average level



2. increases over time

Global average GDP per capita over the long run

These historical estimates of GDP per capita are adjusted for inflation. We combine three sources to create this time series: the Maddison Database (before 1820), the Maddison Project Database (1820-1989), and the World Bank (1990 onward).



Data source: Data compiled from multiple sources by World Bank (2025); Bolt and van Zanden - Maddison Project Database 2023; Maddison Database 2010

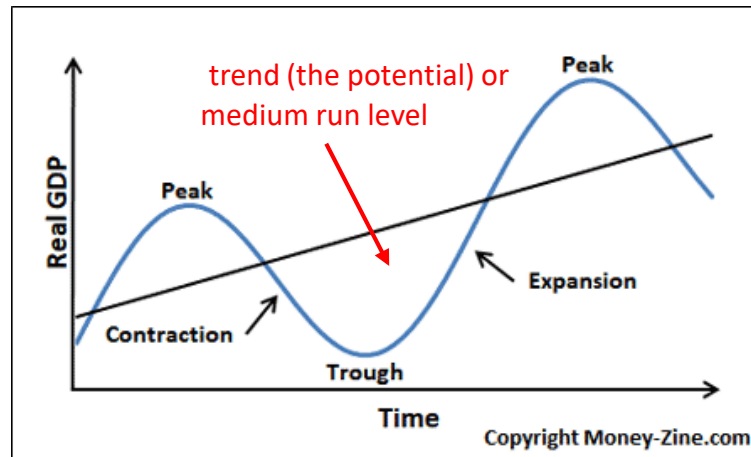
Note: This data is expressed in international-\$' at 2021 prices.

OurWorldinData.org/economic-growth | CC BY

1. **International dollars:** International dollars are a hypothetical currency that is used to make meaningful comparisons of monetary indicators of living standards. Figures expressed in international dollars are adjusted for inflation within countries over time, and for differences in the cost of living between countries. The goal of such adjustments is to provide a unit whose purchasing power is held fixed over time and across countries, such that one international dollar can buy the same quantity and quality of goods and services no matter where or when it is spent. Read more in our article: [What are Purchasing Power Parity adjustments and why do we need them?](#)

THE BUSINESS CYCLE

- Fluctuations occur in the short run
- The average level is reached in the medium run



MAIN DETERMINANTS OF AGGREGATE OUTPUT

The amount of goods and services an economy can produce is determined by different factors in different time horizons:

- **SHORT RUN** (few quarters/some years):
The level of the demand
- **MEDIUM RUN** (a decade or more):
Structural supply factors (population size, features of the labor market, productivity (human and physical, technology)
- **LONG RUN** (a century or more):
Institutional factors (educational system, quality of political institutions, justice ...)



A SHORT-RUN MODEL

to explain income variations in the short run

What is a model?

Why do we need models?

THE MACRO METHOD

- Macroeconomists study questions of interest using a **scientific method**:
 1. Analyze facts
 2. Build a model
 3. Test the model with real data (if necessary refine the model)
 4. Use the model to predict effects
- Models involve **equations**, can be represented **graphically** and can explained in **word**
(...these are the 3 things you have to learn for the exam!!)
- Models have
 - **Endogenous variables**: to be explained with the model (those on the axes in the graph)
 - **Exogenous variables**:
 - taken as given
 - can be changed to make comparative statics analysis

THE GOALS OF THE MACRO MODELS

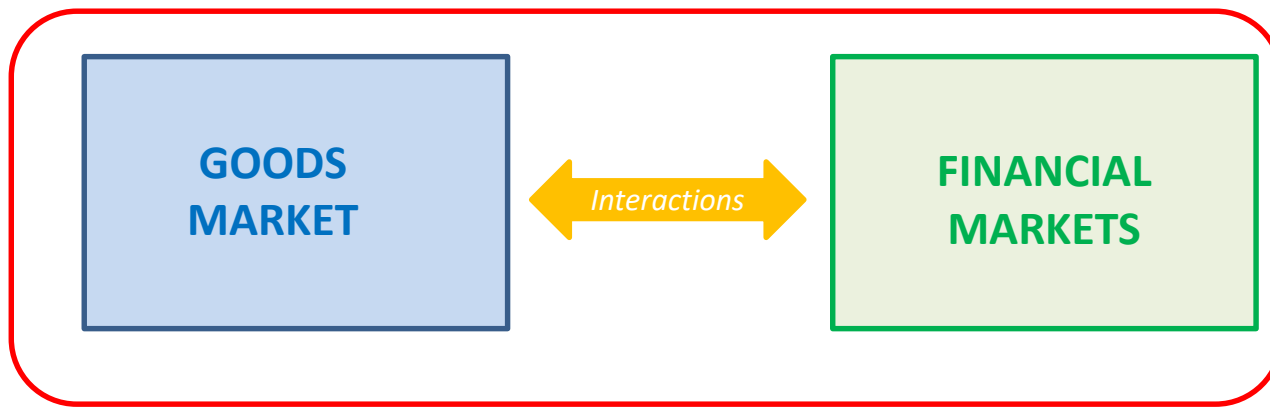
1. Explain reality

- Simplify the complicated real world into its most relevant elements
 - The model does not replicate perfectly every single detail/feature of markets or behavior

2. Make predictions that will eventually be tested

- A model is useful if it has good predictive power
- Politicians and policy makers need to define interventions and policies to face shocks
 - Energy crisis
 - Financial crisis
 - Covid crisis
 - Natural disasters
 - Wars
 - ...

THE MACRO ECONOMY IN A NUTSHELL



The macro dynamics
of a country
(IS-LM model)

THE ROLE OF THE DEMAND IN THE SHORT RUN

- From the national accounting we know that

$$\text{Production (Y)} = \text{Income (Y)} = \text{Spending (Demand) (Z)}$$

⇒ If we demand more firms produce more and vice versa

⇒ If income ↑, demand ↑ and production ↑

In the short run, the amount of goods/services produced is determined by
demand factors

COMPONENTS OF DEMAND

We have to consider who demands a country's goods/services produced by the economy and for what purposes:

- **Consumption (C):**

Goods and services bought by final consumers to satisfy current needs

- **Investment (I):**

Goods and services bought to satisfy future needs
Fixed investment in machinery and residential investment

- **Government expenditure (G):**

Goods and services bought by public sector
Wages of public employees included, transfers (pensions/subsidies) excluded

Domestic demand (DD)
for goods and services

- **Exports (X):**

goods and services produced in the country and consumed by the rest of the world

- **Imports (IM):**

goods consumed in the country but produced abroad

Net Exports = $X - IM$

if $NX > 0 \Leftrightarrow X > IM$
Trade surplus

if $NX < 0 \Leftrightarrow X < IM$
Trade deficit

- **Inventories (In)**

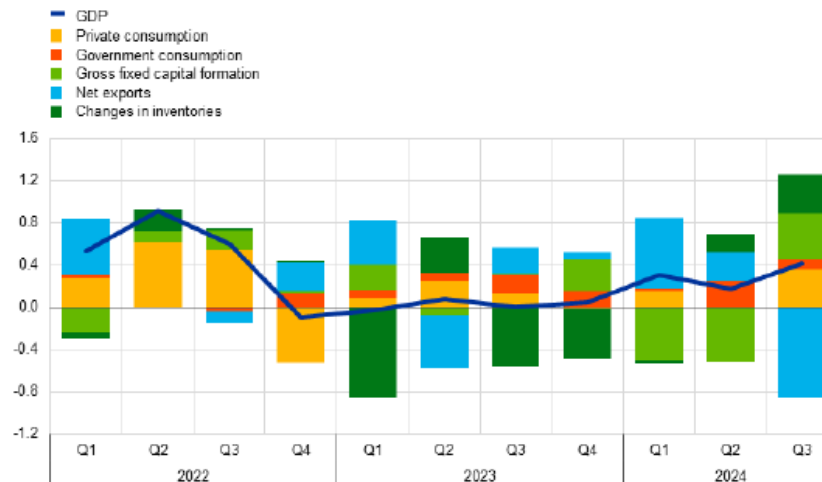
Aggregate demand = Total demand for goods/services produced by the economy

$$Z = C + I + G + X - IM + IN$$

- It is an identity (definition): it is always true

Euro area real GDP and its components

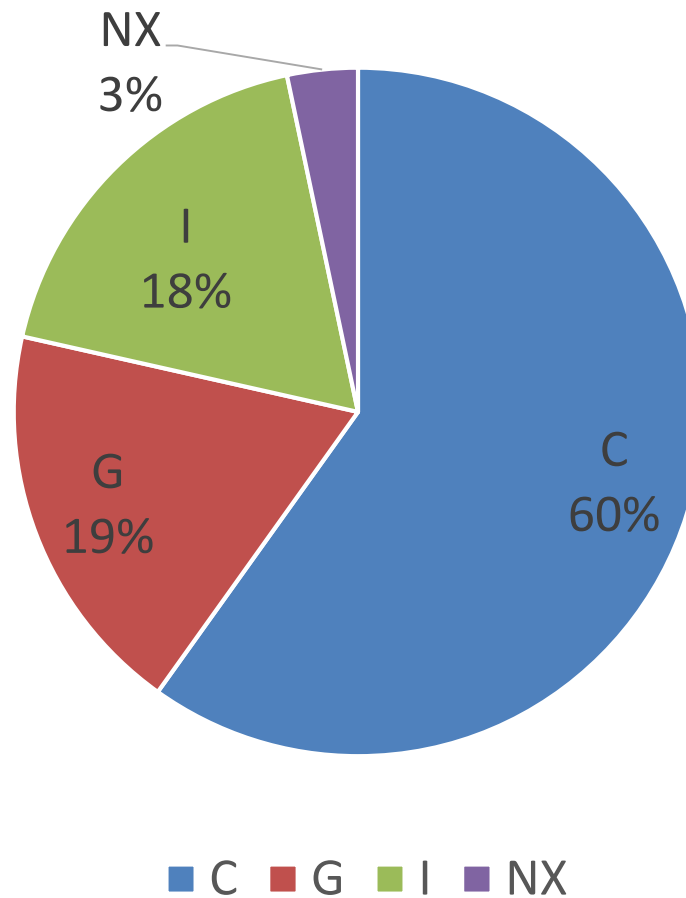
(quarter-on-quarter percentage changes; percentage point contributions)



Sources: Eurostat and ECB calculations.

Note: The latest observations are for the third quarter of 2024.

GDP DECOMPOSITION, Italy



AIM

Explain formally how demand affects production and income



THE GOODS MARKET MODEL IN THE SHORT RUN

ASSUMPTIONS

1. One **single good**: used in consumption and investment
2. In the **short run prices are fixed**: firms can supply any amount at current prices

3. **No inventories**: sales production

In the reality, sales production

- If sales production: inventories
- If sales production: inventories

4. **Closed economy**: autarky (no export nor import)

aggregate demand = demand for domestic goods

$$Z = C + I + G$$

5. **Linear behavioral equations** for aggregates

CONSUMPTION

- How do agents behave? How do they choose?
- What are the determinants of C?

1. Preferences
2. Gross income (received) Y
3. Taxes (paid)
4. Subsidies (received by the public sector)
5.

Disposable income, $Y_D = Y - T$

Gross Income Net transfers
 ↓
 = taxes - subsidies

⇒ If $Y \uparrow$, $Y_D \uparrow$ and $C \uparrow$

⇒ There is a positive relationship btw Y and C

$$C = C(Y_D) = C(Y - T)$$

+ +

Behavioral function of consumption

To solve analytically the model, we may use a specific functional linear form for consumption

$$C = c_0 + c_1(Y - T)$$


Autonomous consumption

- Amount consumed if disposable income is zero
- $c_0 > 0$

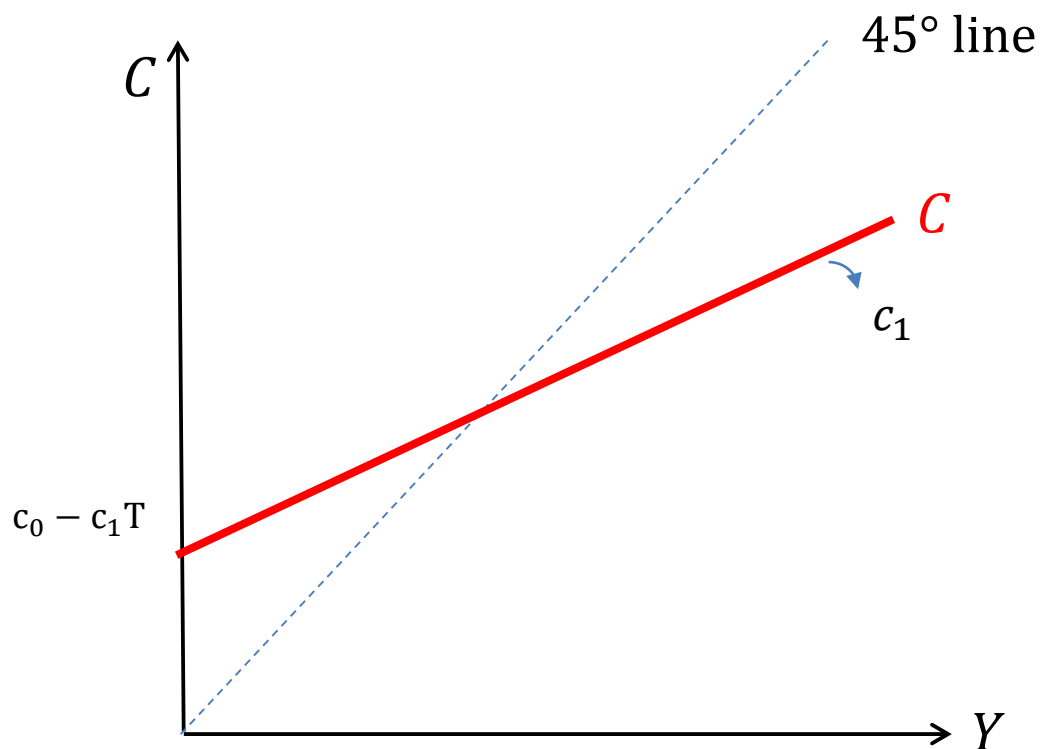
Marginal propensity to consume

- Impact on consumption of an additional € of disposable income
- $0 < c_1 < 1$

Assume $c_1 = 0,6$:
if $Y \uparrow$ by 100, $C \uparrow$ only by 60 =>
40 are saved

CONSUMPTION: graphical representation

$$\begin{aligned} C &= c_0 + c_1(Y - T) \\ &= c_0 - c_1T + c_1Y \end{aligned}$$



OBSERVATIONS:

1. C function is drawn for a given level of T and c_0

⇔ T and c_0 are exogenous variables

2. Since $c_1 < 1$ the C function is flatter than the 45° line

⇔ When $Y \uparrow$, $C \uparrow$ less

⇔ If $Y \uparrow$ by ΔY , $C \uparrow$ by $c_1 \Delta Y$

COMPARATIVE STATICS

What happens if...

...Y increases?

...T increases?

... c_0 increases?

CONSUMPTION: graphical representation

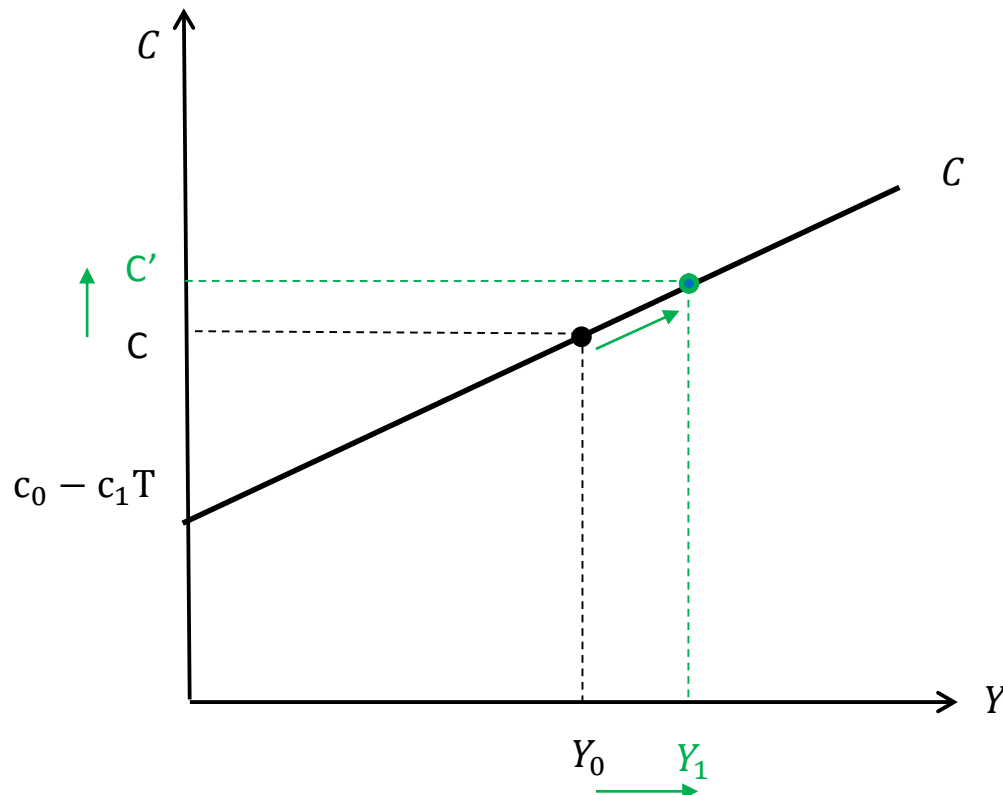
What happens if Y increases?

$$\begin{aligned} C &= c_0 + c_1 (Y - T) \\ &= c_0 - c_1 T + c_1 Y \end{aligned}$$

Y is an endogenous variable
for the C function



Movement ALONG
the curve



CONSUMPTION: graphical representation

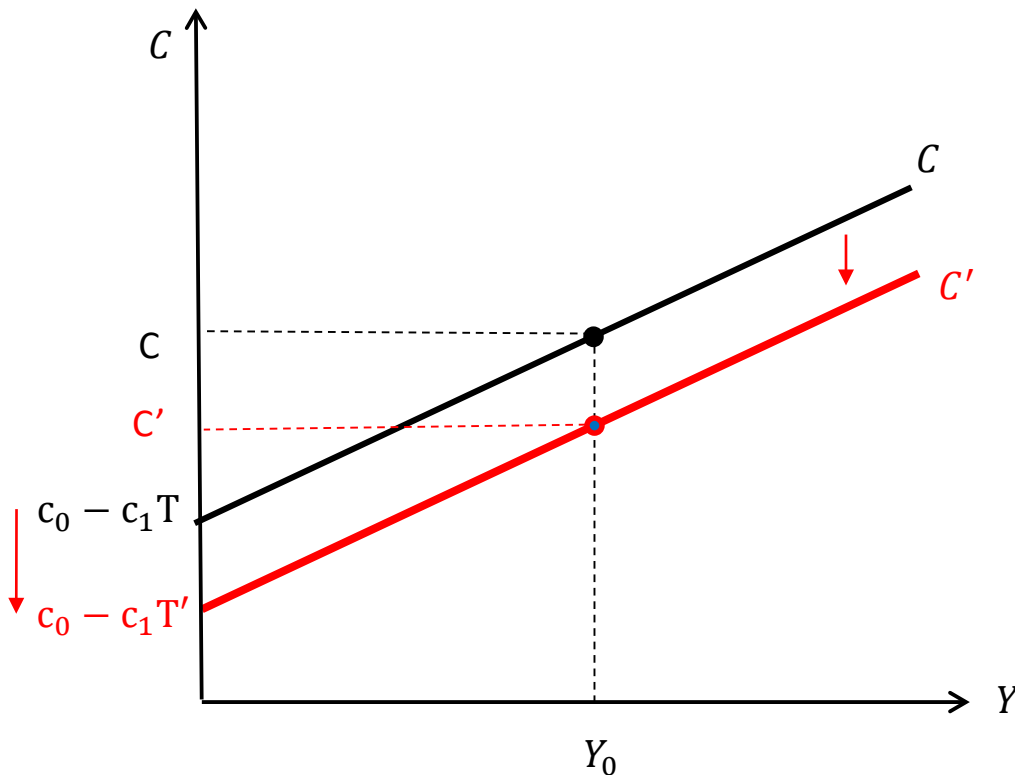
What happens if T increases?

$$\begin{aligned} C &= c_0 + c_1 (Y - T) \\ &= c_0 - c_1 T + c_1 Y \end{aligned}$$

T is an exogenous variable for the C function



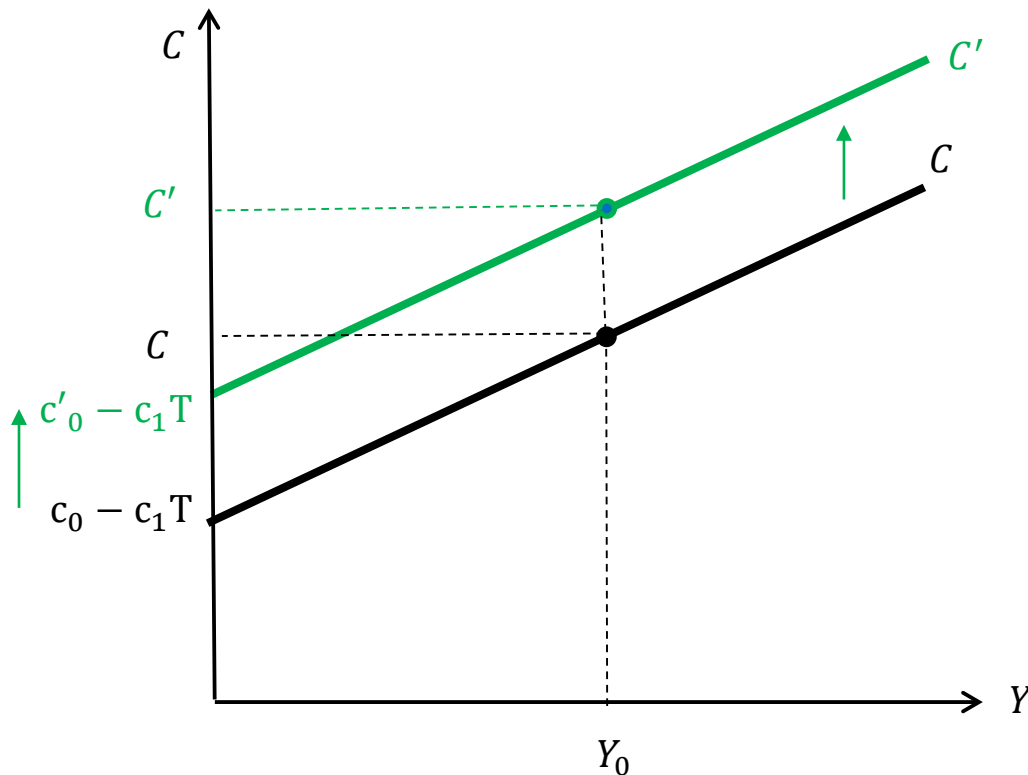
The curve SHIFTS



CONSUMPTION: graphical representation

What happens if c_0 increases?

$$C = c_0 + c_1(Y - T) \\ = c_0 - c_1T + c_1Y$$



c_0 is an exogenous variable
for the C function

↓
The curve SHIFTS

INVESTMENT

- How do firms decide how much to invest in physical capital?
- What are the determinants of I?

We start assuming that investment is **exogenous**

$$I = \bar{I}$$

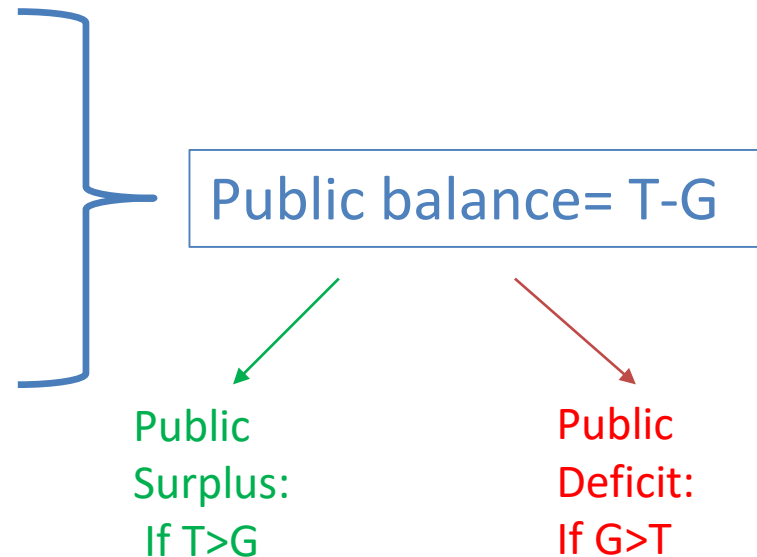
We will remove this assumption later on...!

GOVERNMENT

- What variables are under the control of the government?

- Government expenditure, G

- Net Transfers, T



- We start assuming they are exogenous
- G and T characterize the FISCAL POLICY of the economy

AGGREGATE DEMAND

- Using our assumptions the aggregate demand will be:

$$Z = \underbrace{(c_0 - c_1T + \bar{I} + G)}_{\text{Autonomous spending: A}} + c_1Y$$

Autonomous spending: A

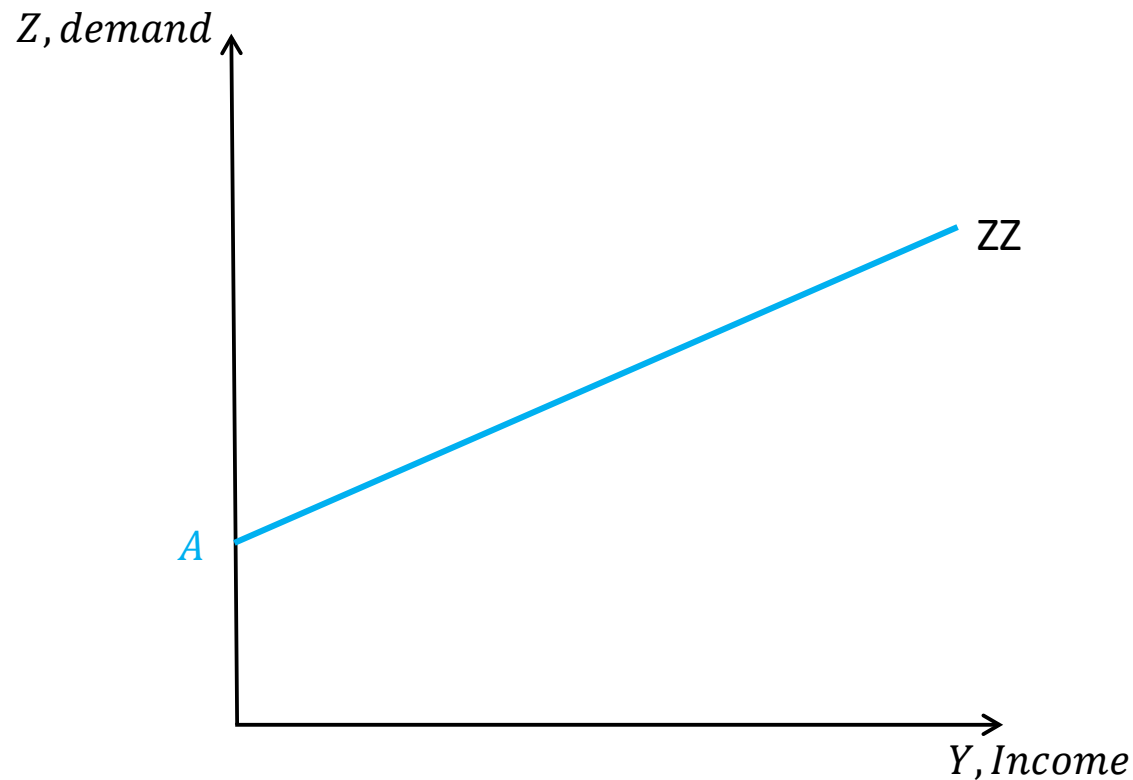


All the components of the D that
do not depend on income

GRAPHICAL REPRESENTATION

- We can represent the aggregate demand (Z) as a function of income (Y)

$$Z = (c_0 - c_1T + \bar{I} + G) + c_1Y$$



GOODS MARKET: THE EQUILIBRIUM

EQUILIBRIUM

The goods market is in equilibrium when the **quantity supplied** is equal to the **quantity demanded**

$$Y = Z$$

No excess demand nor excess supply

HOW TO FIND THE EQUILIBRIUM

We follow a 3-step procedure to find the equilibrium

1. Analytical solution
2. Graphical representation
3. Economic intuition

... your three key skills to perform well in macro!!

ANALYTICAL SOLUTION

Impose the equilibrium condition, plug the behavioral equations and solve for Y

$$\begin{cases} Z = (c_0 - c_1T + \bar{I} + G) + c_1Y \\ Y = Z \end{cases}$$

$$Y = (c_0 - c_1T + \bar{I} + G) + c_1Y$$

$$(1 - c_1)Y = (c_0 - c_1T + \bar{I} + G)$$

$$Y = \frac{1}{1 - c_1} (c_0 - c_1T + \bar{I} + G)$$

Multiplier of the goods
market, m

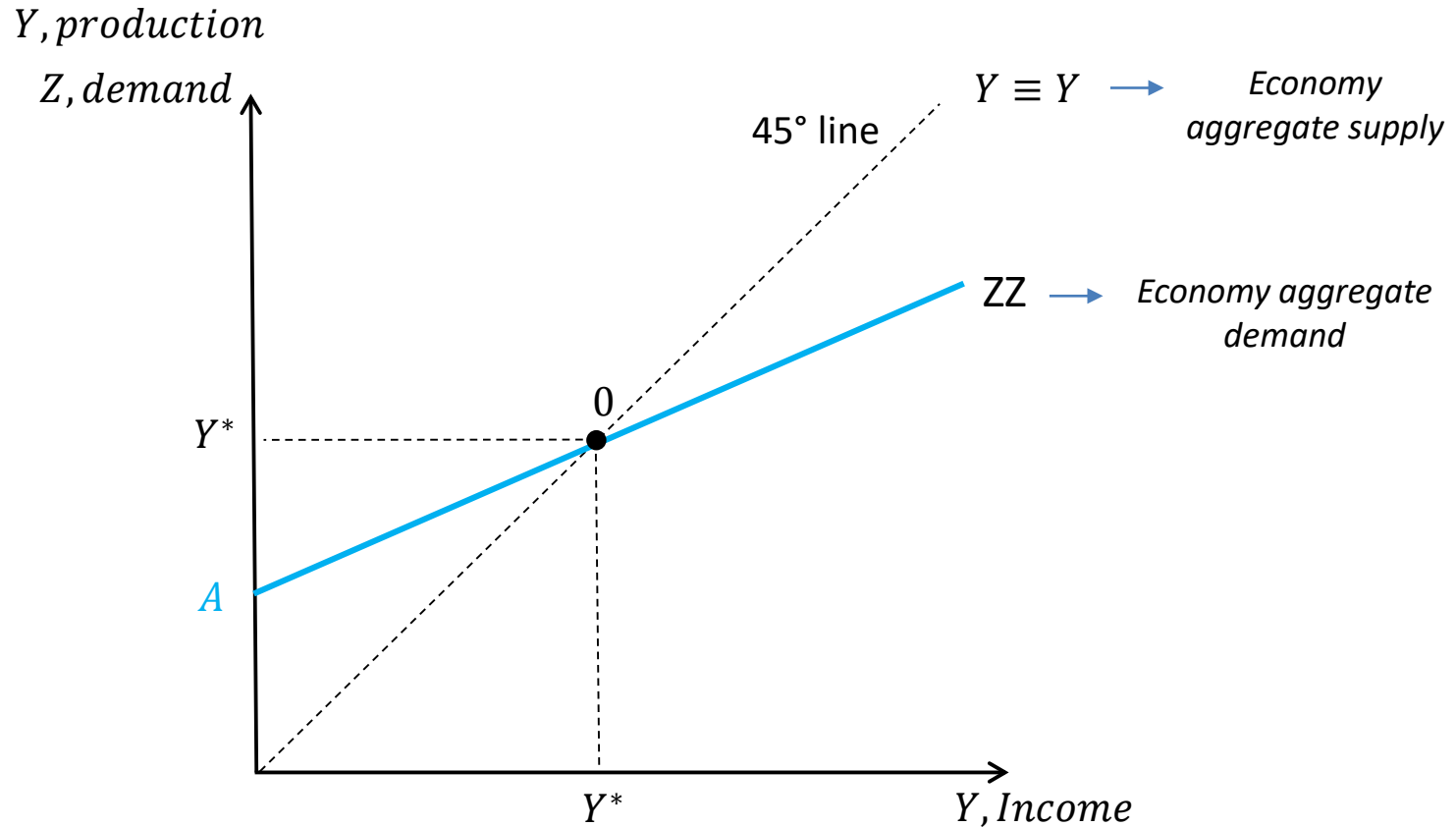
Autonomous spending, A

GRAPHICAL REPRESENTATION

1. Plot demand as a function of income $Z=f(Y)$
2. Plot production as a function of income $Y=f(Y)$
3. Impose the equilibrium condition $Y=Z$

GRAPHICAL REPRESENTATION

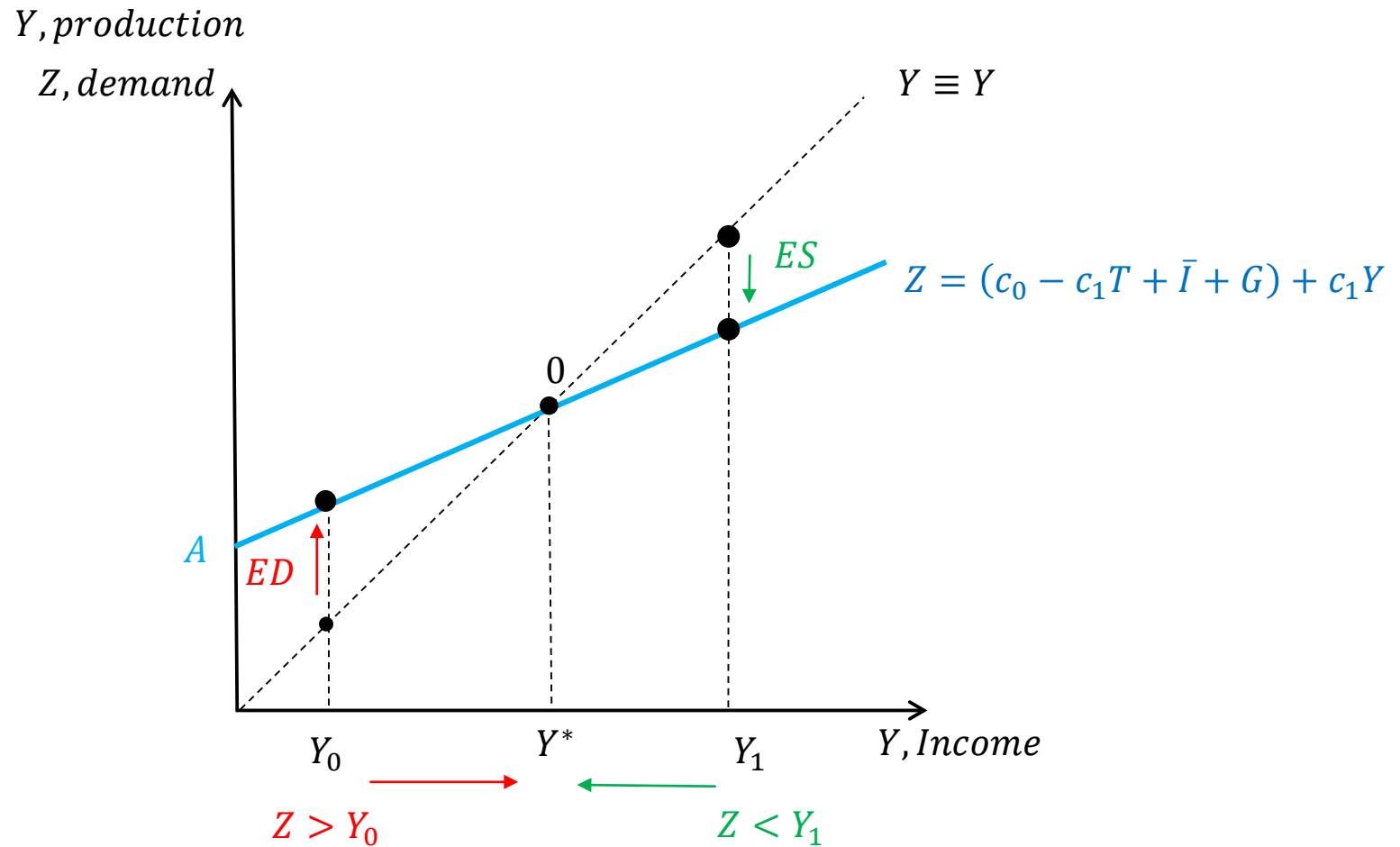
$$Z = (c_0 - c_1T + \bar{I} + G) + c_1Y$$



**WHAT HAPPENS IF THE ECONOMY IS
NOT IN EQUILIBRIUM?**

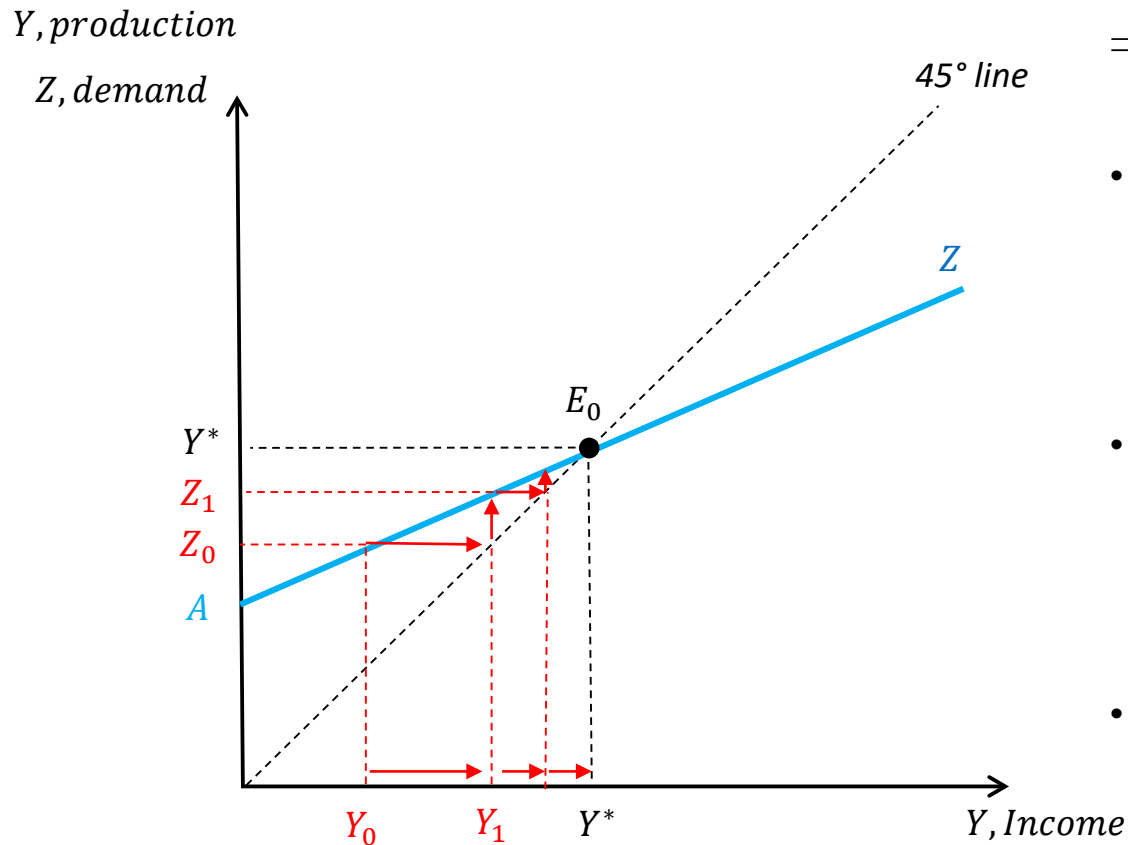
$$Y > Z \quad \text{or} \quad Y < Z$$

EXCESS DEMAND AND SUPPLY



INTUITION

Suppose that income is $Y_0 < Y^*$



- If $Z > Y \Rightarrow$ excess demand
 $\Rightarrow$ Firms produce more
 $\Rightarrow$ Income increases : $Y_0 \rightarrow Y_1$
- As income is higher, consumption and thus demand increases further (Z_1) $\Rightarrow$ production increases more leading to a further increase in $C, Z, Y, C, Z, Y, \dots$
- However, the increase in demand induced by the initial change in income is less than proportional as $c_1 < 1$
- Eventually the system converges to the equilibrium: $Y_0 \rightarrow Y^*$

...make alone the reasoning for the opposite case !!!

USING THE MODEL: COMPARATIVE STATICS

COMPARATIVE STATICS

- Suppose the goods market is **in equilibrium** (point 0)
- Assume that a **shock hits** the goods market or a **policy is introduced**

...we have to

1. Find the **new equilibrium** (point 1)
2. **Compare** the two equilibria:
 - Draw the graph
 - Explain with words
 - Discuss the economic intuition

!!! Remember to follow this multiple step procedure!!!

The UK invests £90 million into their road infrastructure

February 7, 2025



The UK government will invest £90 million into the country's transport infrastructure through four major road schemes. These upgrades are designed to reduce journey times, improve road safety, boost local economies, and make travel more efficient for cars and buses.

The government's investment is part of its commitment to prioritising infrastructure projects that provide value for money, support economic growth, and enhance living standards.

This project will reduce journey times by up to 25%, thanks to the duality of two bypass sections and the roundabout improvements.

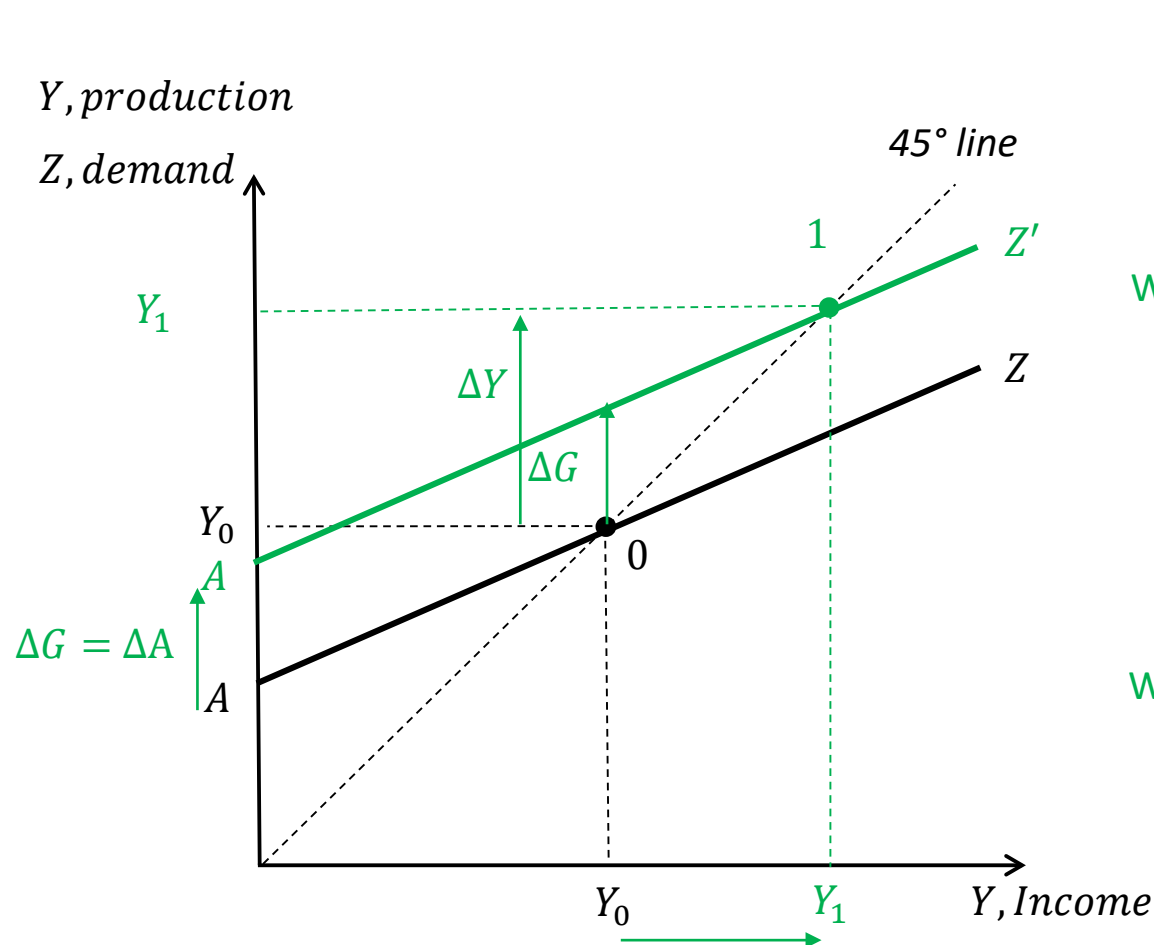
These changes are expected to reduce congestion on local roads and improve safety for both drivers and pedestrians. Businesses will benefit from more efficient transportation, allowing goods to move freely across this critical part of the UK's freight network. As a result, the local economy should experience a boost, with reduced travel times leading to increased productivity.

FISCAL POLICIES

- Fiscal policy is implemented when the government:
 - Changes government expenditure G
 - Changes net transfers (taxes) T
- Expansionary fiscal policy: $G \uparrow$ and/or $T \downarrow$
- Restrictive fiscal policy: $G \downarrow$ and/or $T \uparrow$

- The government constructs new public transportation infrastructures
 - roads and highways, bridges, tunnels, railways, airports, ports harbors...
 - Structures that support transportation and logistics
- $\uparrow G \Leftrightarrow \Delta G > 0 \Leftrightarrow \text{EXPANSIONARY FP}$

1. GRAPHICAL ANALYSIS



$$Y = Y$$

$$Z = (c_0 - c_1T + \bar{I} + G) + c_1Y$$

Which curve is affected?

- G is an **exogenous** determinant of demand
- Z shifts up

What happens to the equilibrium?

The equilibrium moves
from point 0 to 1:
Higher income/production

2. ANALYTICAL SOLUTION

- What is the impact on Y if government expenditure changes G by ΔG ?

ΔY due to ΔG

$$\Delta Y = \frac{1}{1 - c_1} (\Delta c_0 - c_1 \Delta T + \Delta \bar{I} + \Delta G)$$

$$\Delta Y = \frac{1}{1 - c_1} \Delta G > 0$$

$$|\Delta Y| > |\Delta G|$$

As $0 < c_1 < 1, \frac{1}{1 - c_1} > 1 \Rightarrow$ The impact of the increase of G on Y is more than proportional

3. ECONOMIC INTUITION

- Why does a change in G result in a more than proportional change in Y ?

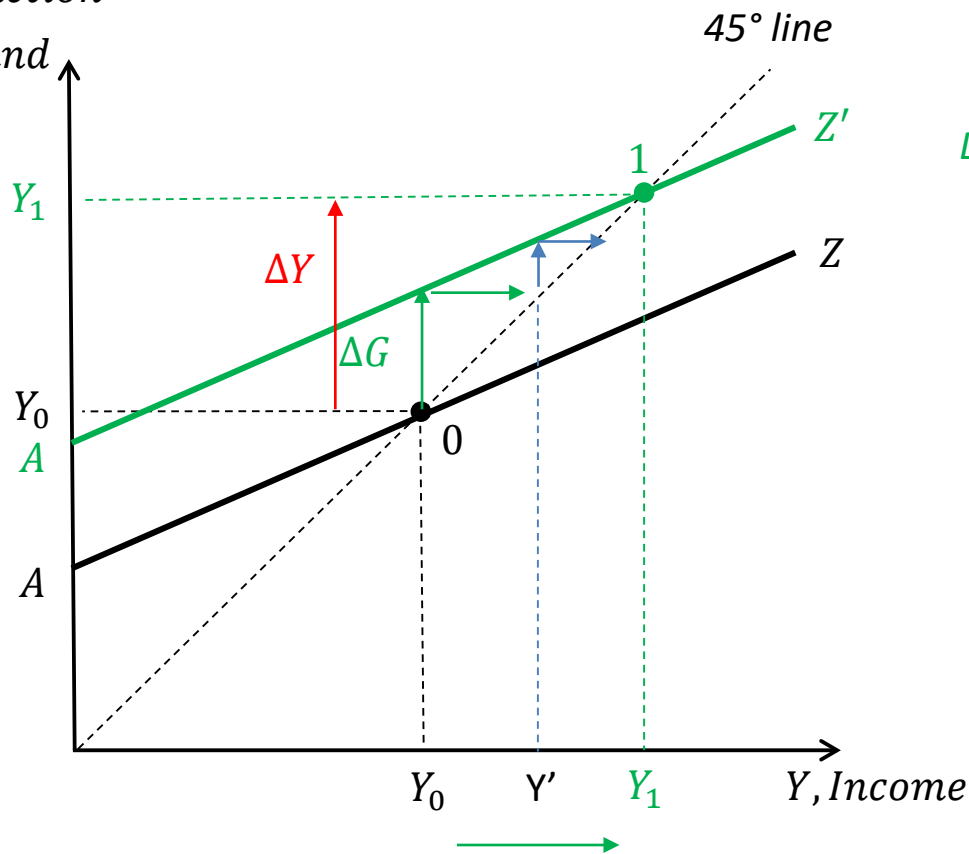
There are some direct and indirect effects

- Direct effect: $\uparrow G = \uparrow Y$
 - The increase in government expenditure raises demand
 - As demand is higher, firms produce more
- Indirect effect : $\uparrow Y \rightarrow \uparrow C \text{ by } c_1 \rightarrow \uparrow Z \rightarrow \uparrow Y \rightarrow \uparrow C \dots$
 - More production = More income
 - If income increases, consumption increases (but less than proportionally)
 - Demand increases production increases...
 - ... demand increases further more...

The final effect (sum of direct and indirect effects) on output is more than proportional

3. INTUITION GRAPHICALLY

Y , production
 Z , demand



$$Y = Y$$

$$Z = (c_0 - c_1T + \bar{I} + G) + c_1Y$$

Direct effect : $Y_0 \rightarrow Y'$
 $G \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow$

Indirect effect
(multiplier) : $Y' \rightarrow Y_1$

$Y \uparrow \Rightarrow C \uparrow \Rightarrow Z \uparrow \Rightarrow Y$
 $\uparrow \Rightarrow \dots Y \uparrow \uparrow$

Timing?

THE MULTIPLIER

- The goods market multiplier captures the sum of all direct and indirect effects
- Given an initial change in the autonomous spending, in equilibrium, the production increases more than proportionally
- It measures by how much income changes if autonomous spending changes by a given amount

EFFECT ON CONSUMPTION

- By how much does C change in the new equilibrium as a result of the change in G?

$$C = c_0 + c_1 (Y - T)$$

$$\Delta C = \Delta c_0 + c_1 (\Delta Y - \Delta T)$$

$$\Delta C = \Delta c_0 + c_1 \Delta Y - c_1 \Delta T$$



Since C depends on Y, the change in Y as a result of the change in G must be considered when quantifying the change in C

$$\Delta C = 0 + c_1 \Delta Y - c_1 0$$

$$\Delta C = c_1 \Delta Y$$

$$\Delta C = c_1 (1/1 - c_1) \Delta G$$

$$\Delta C > 0$$

In the new equilibrium aggregate consumption will be higher

DEMAND SHOCKS

- Other positive demand shocks produce the same qualitative results
 - $I \uparrow$
 - $c_0 \uparrow$ (better confidence of consumers)
 - $c_1 \uparrow$
 - $T \downarrow$
- But depending on the shock the quantitative effect may be different

*Try to repeat the all
reasoning we made alone...*

MAGNITUDE OF THE IMPACTS

A demand shock produces higher effect on income when the multiplier is higher

- The multiplier increases if the marginal propensity to consume increases
- The multiplier increases if the n. of the components of the demand that depend positively on income increases

- Consider two countries

- Country A: consumers save 60% of their income
- Country B: consumers save 80% of their income

Will the multiplier be higher in A or B? Why?

m_A and m_B ?

- Consider two countries

- Country A: only C is endogenous
- Country B: C and I depend positively on income

$$C = c_0 + c_1 Y$$

$$I = I_0 + d_1 Y \text{ where } d_1 \text{ is the marginal propensity to invest}$$

Will the multiplier be higher in A or B? Why?

m_A and m_B ?

INVESTEMENT AND SAVING

PRIVATE SAVING

Portion of disposable income NOT consumed

$$\text{Private Saving} = S_{\text{priv}} = (Y - T) - C$$


$$C = c_0 + c_1(Y - T)$$

Plugging the behavioral equations for C and T we obtain

$$S = (Y - T) - c_0 - c_1(Y - T)$$

$$S = -c_0 + \underbrace{(1 - c_1)}_{\text{Marginal propensity to save}}(Y - T)$$

*Autonomous
saving*

*Marginal
propensity to save*

PUBLIC SAVING

Government saving depends on fiscal policy: it is the difference btw revenues and spending of the public sector

$$\text{Public saving: } S_{\text{Gov}} = T - G$$

If $S_{\text{Gov}} > 0 \Leftrightarrow T > G$:
budget surplus
The public saves

If $S_{\text{Gov}} < 0 \Leftrightarrow T < G$:
budget deficit
The public has to borrow money

NATIONAL SAVING

Total saving in the economy

$$S_{\text{nat}} = S_{\text{Priv}} + S_{\text{gov}}$$
$$S_{\text{nat}} = (Y - T - C) + (T - G)$$

ALTERNATIVE EQUILIBRIUM CONDITION

Let's write in another way the equilibrium condition for the goods market

$$Y = Z$$

$$Y = C + I + G$$

$$Y - T = C + I + G - T$$

rearranging

$$(Y - T - C) + (T - G) = I$$

S_{priv}
Private saving

S_{gov}
Public saving

Production = Demand

*Alternative equivalent
equilibrium conditions*

Total (national) Saving = Investment

OBTAINING THE EQUILIBRIUM

We can find the short run equilibrium for GDP using this alternative equilibrium condition

$$S_{\text{Priv}} + S_{\text{Gov}} = I$$

$$-c_0 + (1 - c_1)(Y - T) + (T - G) = I$$

$$(1 - c_1)Y = c_0 + I + (1 - c_1)T + G - T$$

$$(1 - c_1)Y = c_0 + I + T - c_1T + G - T$$

$$(1 - c_1)Y = c_0 + I - c_1T + G$$

$$Y = \frac{1}{1 - c_1} (c_0 + I - c_1T + G)$$

...as by imposing $Y=Z!!!$

QUICK CHECK

- If in country A the marginal propensity to save is higher than in country B, and autonomous spending is the same in both countries, then the output in the short run will be:
 - a) lower in country A.
 - b) lower in country B.
 - c) equal in both countries.
 - d) more unstable in country A.

QUICK CHECK

- In the goods market model in the short run, a decrease in net taxes
 - a) decreases income and consumption in equilibrium.
 - b) increases income and decreases consumption in equilibrium.
 - c) decreases income and increases consumption in equilibrium.
 - d) increases income and consumption in equilibrium.

QUICK CHECK

- In an economy investment decreases by 100 and net taxes increase by the same amount, the aggregate demand _____ and the short run equilibrium income _____
 - a) does not change; increases
 - b) increases; increases.
 - c) decreases; decreases.
 - d) does not change; does not change.

QUICK CHECK

- In an economy investment increases by 100 and net taxes decrease by the same amount, the aggregate demand _____ and the short run equilibrium income _____
 - a) does not change; increases
 - b) increases; increases.**
 - c) decreases; decreases.
 - d) does not change; does not change.

CASE STUDY: THE COVID-19 AND THE GOODS MARKET

- COVID-19 led to a sudden economic downturn due to lockdowns
- Huge demand shocks impacted the economy
- We can use the model to answer two key questions
 - How did aggregate demand change?
 - How did policies respond?



IMPACT ON AGGREGATE DEMAND COMPONENTS

- $C \downarrow$ due to restrictions, lower income and uncertainty
- $I \downarrow$ as firms postponed spending
 - Stronger impact on some sectors
- X and $IM \downarrow$ due to disruptions in global trade
- Aggregate demand $\downarrow$ $Z = C + I + G + X - IM$
- Fall in output and economic downturns
- Government intervention to stimulate demand and prevent deep recession.
 - Expansionary FP to stabilize Y with
 - $\uparrow G$: more investment in health, investment in infrastructures, spending in education and digitalization, finance vaccine research, production, and free distribution (US, EU, UK, France, Italy, China, India)
 - $\downarrow T$: unemployment benefits, deferred income tax, VAT, and payroll tax payments, cut Value Added Tax rates (Germany, UK, France, Italy)
- Rapid demand increase in 2021-2022

CONCLUSIONS: LESSON FROM COVID-19

- Importance of government intervention in demand shocks
- Role of fiscal policy to ensure economic stability
- Need for balancing recovery with inflation control

OBSERVATIONS: FISCAL POLICY in the REALITY

- Changing or is **not easy** in the reality
- Cutting or increasing can lead to large **budget deficits** and **public debt** in the long run
- **Investment** and imports may change, making it hard for governments to assess the effects of their policies
- **Expectations** are likely to matter
- The effects on output (and inflation) may be unsustainable in the **medium run**

TAKE AWAY AND KEY CONCEPTS

- A short-run model for the goods market
 - Assumptions
 - Behavioral equations
 - Aggregate demand and its components
 - Multiplier and autonomous expenditure
 - Analytical solution
 - Graphical representation
 - Economic intuition
 - Effect of positive and negative demand shocks
- Comparative statics
 - Compare to equilibria after a shock/policy
 - Demand shocks due to
 - Changes in autonomous consumption,
 - Changes in exogenous investment
 - Expansionary vs. Contractionary fiscal policy
- 2 alternative equilibrium conditions
 - $Y=Z$
 - $S_{\text{Priv}} + S_{\text{Pub}} = I$

MAKE PRACTICE

Try alone to say...

... What is the impact of the following shocks/policies on the goods market?

- An increase/decrease in net transfers by 100
- An increase/decrease in autonomous consumption by 10
- An increase/decrease in exogenous investment
- An increase in the marginal propensity to consume

... Make the graph

... derive the change in income and savings

... Explain in words